

Time Series

Applied Time Series Analysis STA457 Lecture Notes

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Time Series Analysis

Parameter Estimation in Time Series Models

Overview

- ➊ In the previous lectures, we learned about how to specify our time series model (decide which specific model to use).
- ➋ The general model we have considered is the ARIMA(p, d, q) model.
- ➌ The simpler models like AR, MA, and ARMA are special cases of this general ARIMA(p, d, q) model.
- ➍ Now assume we have chosen appropriate values of p, d, and q (possibly based on evidence from the ACF, PACF, and/or EACF plots).
- ➎ In the case of nonstationary original data, we can assume that taking d differences has produced differenced data that displays stationarity.
- ➏ Assume that our observed time series data $x_1, \dots, x_n$ follow an ARMA(p, q) model.
- ➐ We now must estimate the unknown parameters in that ARMA(p, q) model:
 - ➊ The method of moments (MOM)
 - ➋ Least squares estimation (LS)
 - ➌ Maximum likelihood estimation (ML)

Method of Moments - Autoregressive Models

- 1 The method of moments is frequently one of the easiest. The method consists of equating sample moments to corresponding theoretical moments and solving the resulting equations to obtain estimates of any unknown parameters.
- 2 Consider the AR(2) model given by

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t.$$

The relationships between the parameters ϕ_1 and ϕ_2 and various moments are given by the Yule-Walker equations:

$$\begin{cases} \rho_1 = \phi_1 + \rho_1 \phi_2 \\ \rho_2 = \rho_1 \phi_1 + \phi_2 \end{cases}$$

- 3 The method of moments replaces ρ_1 by r_1 and ρ_2 by r_2 to obtain

$$\begin{cases} r_1 = \phi_1 + r_1 \phi_2 \\ r_2 = r_1 \phi_1 + \phi_2 \end{cases}$$

- 4 Solving for ϕ_1 and ϕ_2 yields

$$\hat{\phi}_1 = \frac{r_1(1 - r_2)}{1 - r_2^2} \quad \text{and} \quad \hat{\phi}_2 = \frac{r_2 - r_1^2}{1 - r_1^2}$$

Method of Moments - Autoregressive Models (Cont'd)

- 1 For the general AR(p) model, we can similarly write down the Yule-Walker equations:

$$\begin{cases} r_1 = \phi_1 + r_1\phi_2 + r_2\phi_3 + \cdots + r_{p-1}\phi_p \\ r_2 = r_1\phi_1 + \phi_2 + r_1\phi_3 + \cdots + r_{p-2}\phi_p \\ \vdots \\ r_p = r_{p-1}\phi_1 + r_{p-2}\phi_2 + r_{p-3}\phi_3 + \cdots + \phi_p \end{cases}$$

- 2 We can use the following R command to solve this linear system of equations:

`Solve(A, b, fractions=TRUE)`

- 3 The estimates obtained in this way are also called **Yule-Walker estimates**.

Method of Moments - Moving Average Models

- 1 We run into problems when trying to use the method of moments to estimate the parameters of moving average models.

- 2 Consider the simple MA(1) model

$$X_t = W_t - \theta W_{t-1}.$$

- 3 The true lag-1 autocorrelation coefficient in this model is

$$\rho_1 = \frac{-\theta}{1 + \theta^2}.$$

- 4 If we equate ρ_1 to r_1 , we get a quadratic equation in θ .
- 5 If $|r_1| < 0.5$, then only one of the two real solutions satisfies the invertibility condition $|\theta| < 1$.
- 6 That solution is

$$\hat{\theta} = \frac{-1 + \sqrt{1 - 4r_1^2}}{2r_1}.$$

- 7 But if $|r_1| = 0.5$, no invertible solution exists, and if $|r_1| > 0.5$, then no real solution at all exists, and the method of moments fails to give any estimator of θ .

Method of Moments - Moving Average Models (Cont'd)

- ① With higher-order MA(q) models, the set of equations for estimating $\theta_1, \theta_2, \dots, \theta_q$ is highly nonlinear and could only be solved numerically.
- ② For the general MA(q) given by

$$X_t = W_t - \theta_1 W_{t-1} - \theta_2 W_{t-2} - \dots - \theta_q W_{t-q},$$

we have

$$\rho_k = \frac{-\theta_k + \theta_1 \theta_{k+1} + \theta_2 \theta_{k+2} + \dots + \theta_{q-k} \theta_q}{1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2}, \quad k = 1, 2, \dots, q.$$

- ③ When solving this system of nonlinear equations, there would be many solutions, only one of which is invertible.
- ④ In any case, for MA(q) models, the method of moments usually produces poor estimates, so it is not recommended to use the method of moments to estimate MA models.

Method of Moments - Mixed ARMA Models

- 1 Consider only the simplest mixed model, the ARMA(1, 1) model given by

$$X_t = \phi X_{t-1} + W_t - \theta W_{t-1}$$

- 2 We know that

$$\begin{cases} \gamma_0 = \frac{(1-2\phi\theta+\theta^2)\sigma_w^2}{1-\phi^2} \\ \gamma_1 = \phi\gamma_0 - \theta\sigma_w^2 \\ \gamma_2 = \phi\gamma_1 \end{cases}$$

and also

$$\begin{cases} \rho_0 = 1 \\ \rho_1 = \frac{(\phi-\theta)(1-\phi\theta)}{1-2\phi\theta+\theta^2} \\ \rho_2 = \phi\rho_1 \end{cases}$$

- 3 Replacing ρ_1 by r_1 , ρ_2 by r_2 , and solving for ϕ yields

$$\hat{\phi} = \frac{r_2}{r_1}.$$

- 4 To obtain $\hat{\theta}$, the following quadratic equation must be solved and only the invertible solution retained (if any exist):

$$\theta^2(r_1 - \hat{\phi}) + \theta(1 - 2r_1\hat{\phi} + \hat{\phi}^2) + (r_1 - \hat{\phi}) = 0.$$

Method of Moments - Estimation of the Noise Variance

- ④ We should still estimate the variance σ_w^2 of our error component.
- ⑤ For any model, we first estimate the variance of the time series process itself,

$$\gamma_0 = \text{Var}[X_t]$$

by the sample variance

$$s^2 = \frac{1}{n-1} \sum_{t=1}^n (x_t - \bar{x})^2.$$

- ⑥ Then we can take advantage of the known relationships among the parameters in our specified model to obtain a formula for σ_w^2 .
- ⑦ For AR(p) models:

$$\hat{\sigma}_w^2 = (1 - \hat{\phi}_1 r_1 - \hat{\phi}_2 r_2 - \cdots - \hat{\phi}_p r_p) s^2$$

- ⑧ For the AR(1) model, this reduces to $\hat{\sigma}_w^2 = (1 - r_1^2) s^2$.

- ⑨ For MA(q) models:

$$\hat{\sigma}_w^2 = \frac{s^2}{1 + \hat{\theta}_1^2 + \hat{\theta}_2^2 + \cdots + \hat{\theta}_q^2}$$

- ⑩ For ARMA(1,1) models:

$$\hat{\sigma}_w^2 = \frac{1 - \hat{\phi}^2}{1 - 2\hat{\phi}\hat{\theta} + \hat{\theta}^2} s^2$$

Least Squares Estimation

- ➊ Since method of moments performs poorly for some models, we examine another method of parameter estimation: Least Squares.
- ➋ We assume our time series is stationary (or that the time series has been transformed so that the transformed data can be modeled as stationary).
- ➌ To account for the possibility that the mean is nonzero, we subtract μ from each observation and treat μ as a parameter to be estimated.
- ➍ Let us consider a mean-centered AR(1) model:

$$X_t - \mu = \phi(X_{t-1} - \mu) + W_t$$

which can also be rewritten as follows:

$$(X_t - \mu) - \phi(X_{t-1} - \mu) = W_t$$

- ➎ The least squares method seeks the parameter values that minimize the sum of squared differences:

$$S_C(\phi, \mu) = \sum_{t=2}^n [(x_t - \mu) - \phi(x_{t-1} - \mu)]^2$$

- ➏ This criterion is called the *conditional sum-of-squares function* (CSS).

Least Squares Estimates for AR Models

- Now, consider the equation $\frac{\partial S_C}{\partial \mu} = 0$, i.e.,

$$\frac{\partial S_C}{\partial \mu} = 2(\phi - 1) \sum_{t=2}^n [(x_t - \mu) - \phi(x_{t-1} - \mu)] = 0$$

- Solving this equation for μ yields:

$$\hat{\mu} = \frac{1}{(n-1)(1-\phi)} \left[\sum_{t=2}^n x_t - \phi \sum_{t=2}^n x_{t-2} \right]$$

- For large n , we have $\hat{\mu} \approx \bar{x}$, regardless of the value of ϕ .

- Consider now the minimization of $S_C(\phi, \bar{x})$ with respect to ϕ :

$$\frac{\partial S_C}{\partial \phi} = \sum_{t=2}^n (x_t - \bar{x})(x_{t-1} - \bar{x}) - \phi \sum_{t=2}^n (x_{t-1} - \bar{x})^2 = 0$$

- Solving for ϕ gives

$$\hat{\phi} = \frac{\sum_{t=2}^n (x_t - \bar{x})(x_{t-1} - \bar{x})}{\sum_{t=2}^n (x_{t-1} - \bar{x})^2}$$

- This estimator is almost identical to r_1 : it's just missing $(x_n - \bar{x})^2$ in the denominator. So, especially for large n , the least squares and method of moments estimators are nearly identical in the AR(1) model. In the general AR(p) model, the least squares estimators of μ and of $\phi_1, \dots, \phi_p$ are approximately equal to the method of moments estimators.

Least Squares Estimates for MA Models

- 1 Consider now the least squares estimation of θ in the MA(1) model given by

$$X_t = W_t - W_{t-1}$$

- 2 Recall that an invertible MA(1) can be expressed as

$$X_t = W_t - \theta X_{t-1} - \theta^2 X_{t-2} - \theta^3 X_{t-3} - \dots$$

which is an autoregressive model of infinite order.

- 3 In this case

$$S_C(\theta) = [X_t - \theta X_{t-1} - \theta^2 X_{t-2} - \theta^3 X_{t-3} - \dots]^2$$

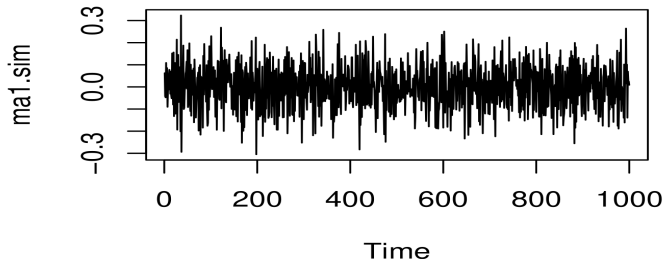
- 4 It is clear from this equation that the least squares problem is nonlinear in the parameters.
- 5 We will not be able to minimize $S_C(\theta)$ by taking a derivative with respect to θ , setting it to zero, and solving. We must resort to techniques of numerical optimization.
- 6 For the simple case of one parameter, we could carry out a grid search over the invertible range $(-1, 1)$ for θ to find the minimum sum of squares. For more general MA(q) models, a numerical optimization algorithm, such as Gauss-Newton will be needed.

Least Squares Estimates for MA Models (Cont'd)

```
set.seed(19)

# simulating MA(1);
ma1.sim<-arima.sim(list(ma = c( -0.2)),
n = 1000, sd=0.1);
```

MA(1), $b = -0.2$, $n=1000$



Least Squares Estimates for MA Models (Cont'd)

```
## SC = conditional sum of squares function for MA(1)

SC<-function(y,theta){
  n<-length(y);
  error<-numeric(n);
  error[1]<-y[1];

  for (i in 2:n){
    error[i]<-y[i]-theta*error[i-1]
  }

  SSE<-sum(error^2)

  return(SSE)
}
```

Least Squares Estimates for MA Models (Cont'd)

```
SC.VEC<-function(y,theta){  
  
    N<-length(theta);  
    SSE.VEC<-numeric(N);  
  
    for (j in 1:N){  
  
        SSE.VEC[j]<-SC(y,theta[j])  
  
    }  
  
    return(SSE.VEC)  
}
```

Least Squares Estimates for MA Models (Cont'd)

```
## Testing function SC

theta=seq(-1,1, by=0.001)

sc.values=SC.VEC(ma1.sim,theta);

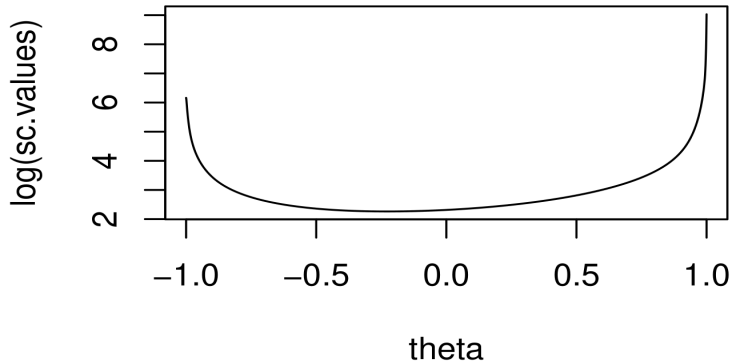
plot(theta,log(sc.values),type="l")

index=seq(1,length(sc.values),by=1);

theta.hat.one=theta[index[sc.values==min(sc.values)]];

theta.hat.one
```


Least Squares Estimates for MA Models (Cont'd)



```
## [1] -0.228
```

Least Squares Estimates for MA Models (Cont'd)

- ① An alternative approach to derive least squares estimate of θ is to proceed by conditioning on one previous value of w_t . Note that $w_t = x_t + \theta w_{t-1}$.
- ② If we set $w_0 = 0$, then we have the set of recursive equations

$$w_1 = x_1$$

$$w_2 = x_2 + \theta w_1$$

$$\vdots$$

$$w_n = x_n + \theta w_{n-1}$$

- ③ Since we know $x_1, x_2, \dots, x_n$ (these are the observed data values) and can calculate the $w_1, w_2, \dots, w_n$ recursively, the only unknown quantity here is θ .
- ④ We can do a numerical search for the value of θ (within the invertible range between -1 and 1) that minimizes $\sum (w_t)^2$, conditional on $w_0 = 0$.
- ⑤ A similar approach works for higher-order MA(q) models, except that we assume $w_0 = w_{-1} = w_{-2} = \dots = w_{-q} = 0$ and the numerical search is multidimensional, since we are estimating $\theta_1, \dots, \theta_q$.

Least Squares Estimates for ARMA Models

- With the ARMA(1, 1) model:

$$X_t = \phi X_{t-1} + W_t - \theta W_{t-1}$$

This can also be rewritten as

$$W_t = X_t - \phi X_{t-1} + \theta W_{t-1}$$

and minimize

$$S_C(\phi, \theta) = \sum_{t=2}^n w_t^2$$

- Note that the sum starts at $t = 2$ to avoid having to choose an initial value x_0 .
- With the general ARMA(p, q) model, the procedure is similar, except that we assume $w_p = w_{p-1} = \dots = w_{p+1-q} = 0$, and we estimate $\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q$.
- For large samples, when the parameter sets yield invertible models, the initial values for $w_p, w_{p-1}, \dots, w_{p+1-q}$ have little effect on the final parameter estimates.

Maximum Likelihood Estimation

- ❶ On the other hand, for small to moderate sample sizes (and for stochastic seasonal models), assuming $w_p = w_{p-1} = \dots = w_{p+1-q} = 0$ can greatly affect the final parameter estimates, which is undesirable.
- ❷ In those cases, rather than using least squares, it may be advantageous to use maximum likelihood estimation.
- ❸ An advantage of maximum likelihood estimation is that it uses all of the information in the data (not just the first few moments as in method of moments).
- ❹ Also, many large-sample results are known about the sampling distribution of maximum likelihood estimators.
- ❺ A disadvantage of maximum likelihood estimation is that we must assume the form of the joint probability distribution of the time series process.
- ❻ The likelihood function is the joint density function of the data, but treated as a function of the unknown parameters, given the observed data $x_1, x_2, \dots, x_n$.
- ❼ For the models we have studied, The likelihood L is a function of the ϕ s, θ s, μ , and σ_w^2 , given the observed $x_1, x_2, \dots, x_n$.
- ❽ The maximum likelihood estimates (MLEs) are the values of the parameters that maximize this likelihood function, i.e., that are the "most likely" parameter values given the data we actually observed.

Maximum Likelihood Estimation for AR(1) Models

- ➊ In the AR(1) model with an unknown but constant mean, the parameters we must estimate are ϕ , μ and σ_w^2 .
- ➋ To perform maximum likelihood estimation in the AR(1) model, we must assume a distribution for our data.
- ➌ The typical assumption is that the $\{W_t\}$ in the AR(1) model are iid $N(0, \sigma_w^2)$ random variables.
- ➍ Under this assumption, the likelihood function is:

$$L(\phi, \mu, \sigma_w^2) = (2\pi\sigma_w^2)^{-n/2} (1 - \phi^2)^{1/2} \exp \left[-\frac{1}{2\sigma_w^2} S(\phi, \mu) \right]$$

where

$$S(\phi, \mu) = (1 - \phi^2)(x_1 - \mu) + \sum_{t=2}^n [(x_t - \mu) - \phi(x_{t-1} - \mu)]^2$$

- ➎ The criterion $S(\phi, \mu)$ is called the *unconditional sum of squares function*.

Maximum Likelihood Estimation for AR(1) Models (Cont'd)

- ④ We must find estimates $\hat{\phi}$, $\hat{\mu}$ and $\hat{\sigma}_w^2$ that maximize the likelihood function (in practice, we typically maximize the log-likelihood function, which produces equivalent estimates).

- ⑤ The estimator of the noise variance σ_w^2 , in terms of the other estimates, is

$$\hat{\sigma}_w^2 = \frac{S(\hat{\phi}, \hat{\mu})}{n}$$

- ⑥ We still need to estimate ϕ and μ .
- ⑦ Comparing the unconditional sum of squares function to the conditional sum of squares function we saw earlier, note that

$$S(\phi, \mu) = S_C(\phi, \mu) + (1 - \phi^2)(x_1 - \mu)^2,$$

so for large sample sizes

$$S(\phi, \mu) \approx S_C(\phi, \mu)$$

- ⑧ This implies that our maximum likelihood estimates of ϕ and μ will be very similar to the least squares estimates, at least for large sample sizes.
- ⑨ The likelihood function for general ARMA models is more complicated, but maximum likelihood estimates can usually be found in these models. In practice, for AR models, MA models, general ARMA or ARIMA models, we can often find either the least squares estimates or the maximum likelihood estimates easily using R.

Properties of the Estimators

- ➊ Recall that LS estimators and ML estimators become approximately equal for large samples.
- ➋ So the large-sample properties of LS estimators and ML estimators are identical for basic ARMA-type models.
- ➌ For large n , these estimators are approximately unbiased and normally distributed.
- ➍ Note: For AR models, MOM estimators have identical large-sample properties as LS and ML estimators.
- ➎ But for models with MA terms, MOM estimators have poor performance and should not be used!
- ➏ For example, for the AR(1) model

$$\text{Var}[\hat{\phi}] \approx \frac{1 - \phi^2}{n}$$

and for the MA(1) model

$$\text{Var}[\hat{\theta}] \approx \frac{1 - \theta^2}{n}$$

- ➐ Clearly, the variance of the estimator decreases (i.e., the precision improves) as n increases. For the AR(1) model, the variance of the estimator $\hat{\phi}$ will be low when the true ϕ is near 1. For the MA(1) model, the variance of the estimator $\hat{\theta}$ will be low when the true θ is near 1.

Large-sample Inference about the Model Parameters

- ➊ When the model parameters are estimated by the ML method, then the ML estimators are approximately normally distributed when n is large.
- ➋ So we can use normal-based inference to get, say, confidence intervals for the true values of the parameters.
- ➌ For large samples, a $(1 - \alpha)100\%$ CI for a parameter takes the form:

$$\text{estimate} \pm z_{\alpha/2}(\text{estimated standard error})$$

- ➍ In an AR(1) model, a 95% CI for ϕ is:

$$\hat{\phi} \pm 1.96\sqrt{(1 - \hat{\phi}^2)/n}$$

- ➎ In an MA(1) model, a 90% CI for θ is:

$$\hat{\theta} \pm 1.645\sqrt{(1 - \hat{\theta}^2)/n}$$

- ➏ The ML estimators are not necessarily approximately normally distributed when n is small.
- ➐ So when n is small, we can use a more general approach, bootstrap-based inference, to get confidence intervals for the true values of the parameters.